NareshKumar D

Data Scientist II

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Data Scientist-II at Carbynetech India Pvt. Ltd | Best Paper awardee @ ICInPro-2019 | NIT-Trichy'2020

Publications

1. **A Semantic-Aware Strategy for Automatic Speech Recognition incorporating Deep Learning Models**. *Proceedings of Intelligent System Design* (2021).
2. **A Novel Hybridized Strategy for Machine Translation of Indian Languages**. *ICSCSP, AISC (SCOPUS)* (2020).
3. **A Novel Semantic Approach for Intelligent Response Generation using Emotion Detection Incorporating NPMI Measure**. *Procedia Computer Science Journal.* **(Best Research Award)** (2020).
4. **A Semantic Approach for Entity Linking by Diverse Knowledge Integration incorporating Role-Based Chunking**. *Procedia Computer Science Journal* (2020).
5. **OntoQuest: An Ontological Strategy for Automatic Question Generation for e-assessment using Static and Dynamic Knowledge**. *Fifteenth International Conference on Information Processing.* **(Best Paper Award)** (ICInPro-2019).
6. **A Novel Approach for Inter-Domain Personalized Search based on Semantic Set Expansion** (ICInPro-2019).

Education

Bachelor of Technology, National Institute of Technology, Tiruchirappalli, CGPA: 7.7/10.0 **July 2016 — July 2020**

Courses taken : Data Structures and Algorithms, DBMS, Computer Architecture, Network Security

Online Courses : Deep Learning Specialization (Coursera) by deeplearning.ai

Work Experience

Data Scientist-II, Carbynetech India Pvt. Ltd. **Feb 2021 — present**

1. Custom Object detection model for PID symbol detection using GANs, YOLO v4, detectron2.

- Implemented a custom **Faster RCNN** model for P & ID symbol detection that competes with Template Matching in terms of inference time and mAP.
- The proposed model reduced inference time by **over 50%** as compared to Template Matching while maintaining **94 % mAP.**
- Developed GANs, Image Processing techniques for data balancing. Implemented object detection model incorporating slicing technique for **34 classes**.

2. Quality Inspection of export pallets using Computer Vision.

- Implemented **Azure Custom Vision** model for detection of attributes such as damage, labels, bar-codes and logos on export pallets.
- Integrated **custom python script for object counting**, and other requirements into Azure Custom Vision model.
- Developed end-to-end **web-application using Angular Framework, utilizing Azure App Services**, Blob Storage, and Flask APIs.

3. Trade Promotion Optimization using Machine Learning models, including FbProphet for Sales Forecasting, Ensemble Learning.

- Fine-tuned **FbProphet** model in extending patterns from past data to future dates to perform better in comparison with LSTM, SARIMAX. Consolidated **Seasonality** and other temporal effects.
- **Increased ROI by 40%** by integrating **ExtraTree Regressors, R square: 0.98,** for sales lift calculation.
- Proposed an alternative method for promotion optimization incorporating **Linear Mixed Effects Models and Bayesian Optimization to maximize total sales, ROI** while minimizing promotional costs.

Technologies: Azure App Services, Azure data bricks and Blob storage, Microsoft Azure cloud services, Azure Custom Vision.

Applied Deep Learning Engineer, AIDesign Pvt. Ltd. **July 2020 — Jan 2021**

- Developed end-to-end Deep Learning models to **solve real-world fluid flow and CFD problems** at a significantly faster rate compared to commercial software.
- Developed a **DNN** that predicts desired parameters on several engineering and mathematical problems with **less than 5% relative error** on loading conditions up to **30 times (30x)** those seen in training.
- Developed a CNN model to predict temperature contours with less than 10% relative error for 2-D heat conduction problems on materials and domain sizes not seen in training.

Academic Projects

Research Assistant, Department of Computer Science and Engineering, NIT Trichy. **Mar'19 — May'19**

Emoji Prediction from Twitter data using Machine Learning techniques.

- A text classification problem, mapping input text to their most likely accompanying emoji, utilizing close to 100,000 scraped Twitter tweets for our training set.
- **Stochastic Gradient Classifier**, Multinomial Naive Bayes Classifier, Gaussian Naive Bayes Classifier and Random Forest Classifier have been employed for the implementation.
- It was observed that the Random Forest Classifier gives the best **accuracy of 56% with a runtime of 35.5 seconds**.

Awards and Honors

- **Best Paper Award**, International Conference on Information Processing, ICInPro, ID: 41.
- Honorary **Rosalind membership at London Journal Press** for published work in Procedia Computer Science. Membership ID #CS52299.
- **Best Research Award** by ScienceFather.Inc, Award ID: 1972

Technical Skills and Certificates

Languages & API : Python, C++, R, Flask
Concepts/Frameworks : ML, DL, RL, CNN, RNN, Tensorflow, Keras, NLP, ARIMA, PyTorch
Packages : Scikit-Learn, Numpy, Pandas, NLTK, Scipy, Matplotlib, Statsmodels, Seaborn
Databases & Systems : SQL, Jupyter & Colab Notebooks, AutoEncoders
Certificates : Deep Learning Specialization, DataStructures and Algorithms Specialization
Deployment : Azure App Services, Azure cognitive services, Azure Blob storage, Azure Data Bricks